

Algorithm to solve Feedback numerical example :-

Step 1:- Identify the input and output terminals; is it a series connection or shunt connection. $\rightarrow$ Then you can identify the Amplifier type.

- If the o/p port's connection is in shunt mode; then you are picking the entire output voltage
- If the o/p port's connection is in Series mode, you are picking the entire output current.
- If the I/p port's connection is in shunt mode, you are creating a feedback current path to the I/p.
- If the I/p port's connection is in series mode, you are giving a feedback voltage to the I/p.

I/p	O/p	Amp type
Voltage	Voltage	Voltage Amplifier.
Voltage	Current	Trans Conductance
Current	Voltage	Trans resistance
Current	Current	Current Amplifier.

If the I/p and o/p signal type varies then the word "Trans" Comes.

$$\text{"Trans"} + \frac{\text{o/p}}{\text{I/p}}$$

Step 2:- Identify the feedback network block. Sometime β can be different from the observation [Tutorial 4 $\rightarrow$ problem 3].

Method to Calculate β

(a) Consider the feedback network as a two port network.

[Keep the current dirⁿ as it is.]



- (b) Generate the I/p, o/p matrix format based on the amplifier type. Along with this also calculate loading effect resistances for the input and o/p ports.

voltage Amp



$$I_1 = -V_1 + -I_2$$

$$V_2 = -V_1 + -I_2$$

✓ → Independent

□ → Dependent.

The Feedback network here is converting V_1 to V_2 ; so β is a factor by which this conversion happen.

$$I_1 = G_1 V_1 + -I_2$$

$$V_2 = \beta V_1 + R_2 I_2$$

$$\beta = \left. \frac{V_2}{V_1} \right|_{I_2=0}$$

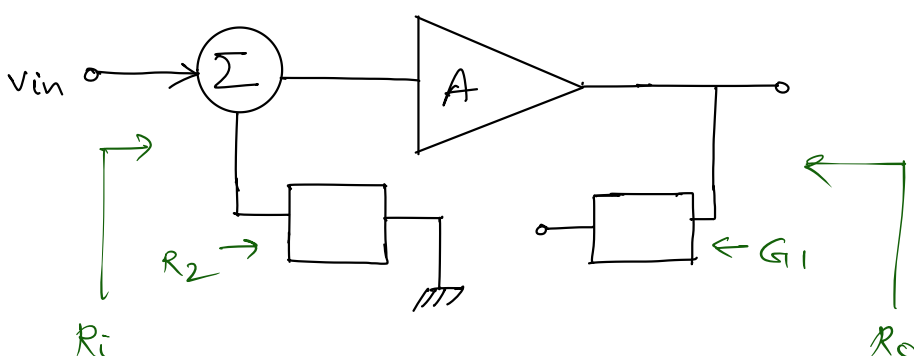
⇒ That means you open the port 2 of the feedback network and see through port 1.

$$G_1 = \left. \frac{I_1}{V_1} \right|_{I_2=0}$$

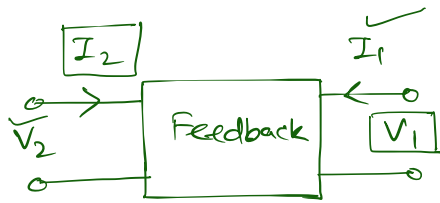
→ Conductance seen through port 1 of the feedback network, when port 2 is open and connect it to the o/p port of the open-loop circuit.

$$R_2 = \left. \frac{V_2}{I_2} \right|_{V_1=0}$$

→ Resistance seen through port 2 of the feedback network, when port 1 is shorted and connect it to the i/p port of the open-loop circuit.



Transconductance Amp



- I_1 you are picking from the o/p port of the main circuit.

Thus, I_1 is independent,
 V_1 is dependent

- V_2 ; you want to deliver as a feedback voltage.

$V_2 \rightarrow$ dependent

$I_2 \rightarrow$ independent

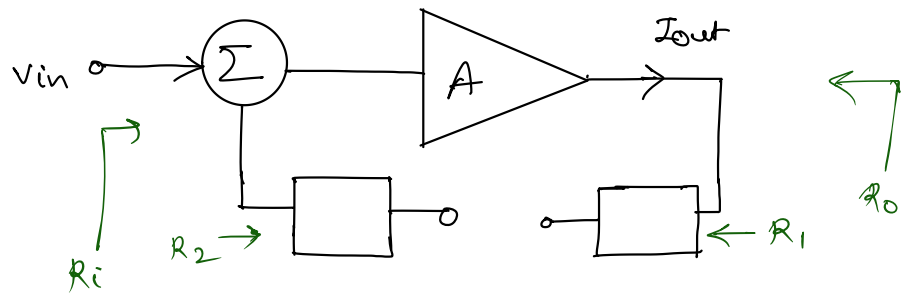
$$V_1 = \underline{R_1} I_1 + \underline{\quad} I_2$$

$$V_2 = -I_1 + \underline{R_2} I_2$$

- From o/p port of the main circuit, you are picking the current I_1 and converting to a voltage V_2

$$\beta = \frac{V_2}{I_1} \Big|_{I_2=0} \quad \text{Open port 2}$$

$$R_1 = \frac{V_1}{I_1} \Big|_{I_2=0} \quad R_2 = \frac{V_2}{I_2} \Big|_{I_1=0}$$



Transresistance Amp



- From o/p port of the main circuit, you are picking V_1
 $V_1 \rightarrow$ independent; $I_1 \rightarrow$ dependent

- At the input port of the main circuit, you are delivering the current I_2

$I_2 \rightarrow$ dependent.

$V_2 \rightarrow$ Independent.

$$I_1 = \underline{G_1} V_1 + \underline{\quad} V_2$$

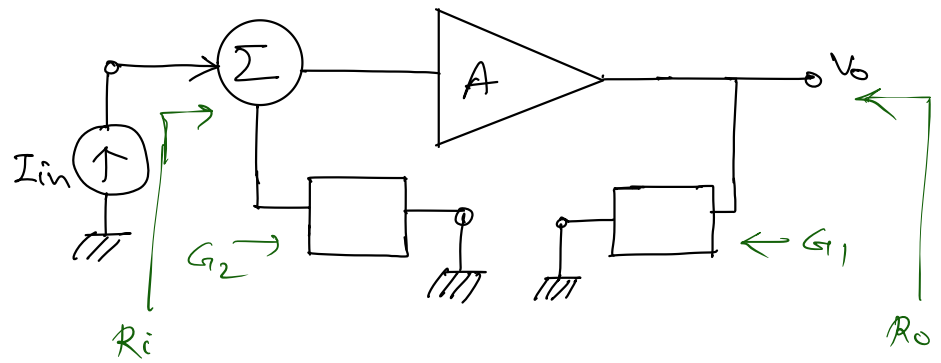
$$I_2 = \underline{\beta} V_1 + \underline{G_2} V_2$$

- Feedback network is taking V_1 and converting to I_2 .

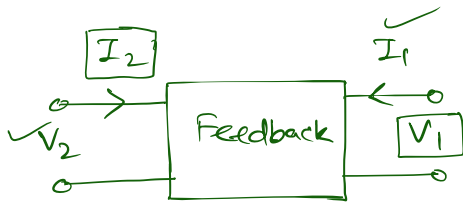
$$\beta = \frac{I_2}{V_1} \Big|_{V_2=0}$$

$$G_1 = \frac{I_1}{V_1} \Big|_{V_2=0} \rightarrow \text{Connect to o/p port of the main Amp.}$$

$$G_2 = \frac{I_2}{V_2} \Big|_{V_1=0} \rightarrow \text{Connect to I/p port of the main Amp.}$$



Current Amplifier



$$V_1 = R_1 I_1 + \frac{1}{\beta} V_2$$

$$I_2 = \frac{1}{\beta} I_1 + G_2 V_2$$

- You are picking the o/p current and supplying the entire current to feedback network as I_1
 $I_1 \rightarrow$ independent ; $V_1 \rightarrow$ dependent

- At the input side, you are feeding back the current I_2 ; which you want to know.

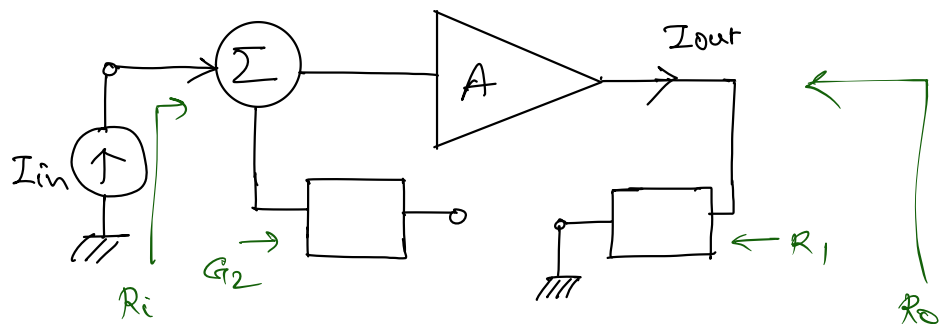
$I_2 \rightarrow$ dependent
 $V_2 \rightarrow$ independent

- Feedback network is taking I_1 and converting to I_2

$$\therefore \beta = \frac{I_2}{I_1} \Big|_{V_2=0}$$

$$R_1 = \frac{V_1}{I_1} \Big|_{V_2=0} \rightarrow \text{Connect to O/p port of the main circuit.}$$

$$G_2 = \frac{I_2}{V_2} \Big|_{I_1=0} \rightarrow \text{Connect to I/p port of the main circuit}$$



- The independent voltage/current source sees a resistance.

$\rightarrow$ If your circuit is open (i.e. A-circuit), then it is R_i

$\rightarrow$ If " " " closed loop then it is R_{if} .

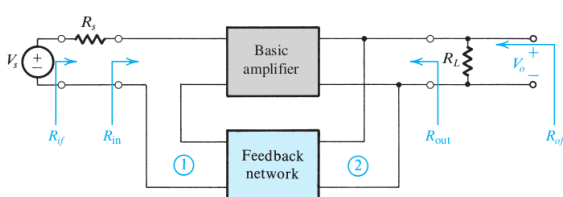
- If the input port is connected in shunt mode; the closed loop resistance will be less; i.e. $R_{if} = \frac{R_i}{1 + AB}$.

- If the input port is Connected in series mode, the closed loop resistance will be more $\therefore R_{if} = R_i (1 + AB)$
- Resistance at the output port, is denoted by
 - $\hookrightarrow$ if Open-loop, R_o
 - $\hookrightarrow$ if closed-loop, R_{of}

Similarly, if the feedback connection at the output port is in shunt mode, the resistance will be less $R_{of} = \frac{R_o}{1 + AB}$.

if the feedback connection at the output is in series mode, the resistance will be more, $R_{of} = R_o (1 + AB)$.

- Generally, R_{in} and R_{out} are provided in the circuit diagram.
- $\rightarrow$ Mostly, R_{in} is the input resistance excluding R_s (voltage or current source's internal resistance).
- $\rightarrow$ R_{out} is the output resistance excluding R_L .

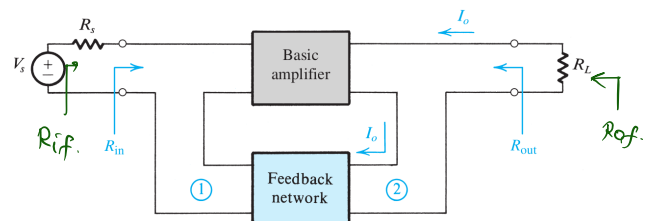


voltage Amp.

$$\begin{aligned} R_{if} &= R_s + R_{in} \\ R_{in} &= R_{if} - R_s \end{aligned}$$

$$\frac{1}{R_{of}} = \frac{1}{R_L} + \frac{1}{R_{out}}$$

$$R_{out} = \frac{1}{\frac{1}{R_{of}} - \frac{1}{R_L}}$$



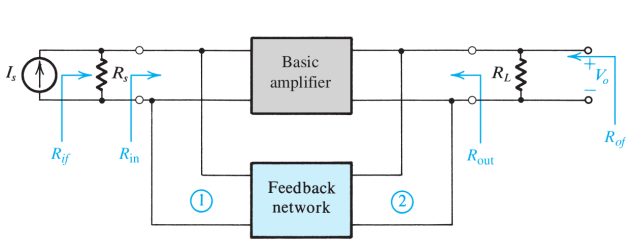
Trans Conductance Amp

$$\begin{aligned} R_{if} &= R_s + R_{in} \\ R_{in} &= R_{if} - R_s \end{aligned}$$

$$\frac{1}{R_{of}} = \frac{1}{R_L} + \frac{1}{R_{out}}$$

$$\frac{1}{R_{out}} = \frac{1}{R_{of}} - \frac{1}{R_L}$$

$$R_{out} = \frac{1}{\frac{1}{R_{of}} - \frac{1}{R_L}}$$



Transresistance Amp.

$$\frac{1}{R_{if}} = \frac{1}{R_s} + \frac{1}{R_{in}}$$

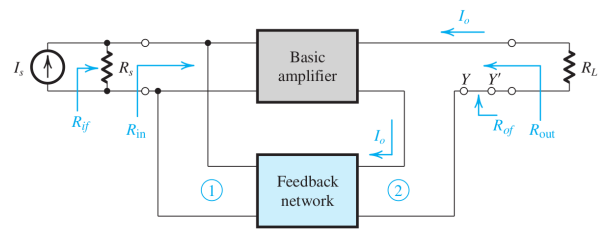
$$\frac{1}{R_{in}} = \frac{1}{R_{if}} - \frac{1}{R_s}$$

$$R_{in} = \frac{1}{\frac{1}{R_{if}} - \frac{1}{R_s}}$$

$$\frac{1}{R_{of}} = \frac{1}{R_L} + \frac{1}{R_{out}}$$

$$\frac{1}{R_{out}} = \frac{1}{R_{of}} - \frac{1}{R_L}$$

$$R_{out} = \frac{1}{\frac{1}{R_{of}} - \frac{1}{R_L}}$$



Current Amplifier

$$\frac{1}{R_{if}} = \frac{1}{R_s} + \frac{1}{R_{in}}$$

$$\frac{1}{R_{in}} = \frac{1}{R_{if}} - \frac{1}{R_s}$$

$$R_{in} = \frac{1}{\frac{1}{R_{if}} - \frac{1}{R_s}}$$

$$R_{of} = R_L + R_{out}$$

$$R_{out} = R_{of} - R_L$$

(*) The name R_i , R_{in} or R_o , R_{out} are given different symbolical name, just use the Concept.